

Ammar Yasser Abdallah

AI & MACHINE LEARNING ENGINEER

Alexandria, Egypt · [+20 1000463804](tel:+201000463804) · ay109543@gmail.com · linkedin.com/in/ammar-yasser2003 · github.com/AmmarYasser72

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer specializing in applied machine learning, deep learning, and (LLM) systems, with hands on experience across NLP/RAG pipelines, computer vision, robotics, and data science. Proficient in Python, scikit-learn, Pandas, and NumPy, with production-style work in LangChain, LangGraph, and Retrieval Augmented Generation (RAG) architectures built on ChromaDB, BM25, and hybrid reranking. Builds complete AI products from signal processing and model development to full stack deployment with React, Next.js, and Docker. Recent Computer & Data Science graduate with a track record of shipping functional AI applications, automation pipelines, and intelligent systems from concept to demo.

TECHNICAL SKILLS

AI & Machine Learning: Python, scikit-learn, Pandas, NumPy, Decision Trees, Random Forest, XGBoost, Model Evaluation (Accuracy, F1-Score, AUC, Cohen's Kappa, MAE)

LLMs, NLP & RAG Systems: Large Language Models (LLMs), LangChain, LangGraph, Google Gemini API, Retrieval-Augmented Generation (RAG), ChromaDB, BM25, Reciprocal Rank Fusion, Transformer/RNN Concepts, Tokenization, BLEU Evaluation

Computer Vision & Signal Processing: OpenCV, MediaPipe, YOLOv8, HRV/PPG Signal Analysis, Feature Extraction, Real-Time Detection Pipelines

Robotics & IoT: Arduino/C++, ESP32, Raspberry Pi, Forward/Inverse Kinematics, Streamlit Dashboards

Full-Stack Development: React, Next.js, TypeScript, Node.js, NestJS, MongoDB, Socket.IO

Tools & Platforms: Docker, Git, GitHub, Postman, n8n (Workflow Automation)

PROFESSIONAL EXPERIENCE

Coding Instructor

Feb 2025 – Mar 2026

O-Academy — Remote

- Delivered live, project-based coding instruction in AI foundations, Python, Scratch, and automation to U.S.-based school-age students, translating complex concepts into accessible, hands-on lessons.
- Designed a progressive curriculum that built students' logic, debugging, and computational-thinking skills through beginner Scratch games, Python exercises, and applied AI activities.
- Adapted lesson plans and pacing across a wide range of ages and skill levels — including extending advanced learners into full-stack fundamentals such as APIs and application logic — sustaining engagement and driving measurable gains in student confidence.

Full Stack Developer Intern

Sep 2025 – Nov 2025

MAIM Digital Solutions — Alexandria, Egypt

- Built responsive full-stack web applications using React.js, Node.js, Express.js, and MongoDB, delivering features from database design through to UI.
- Engineered reusable, user-friendly UI components and integrated REST APIs, partnering closely with backend engineers to ship end-to-end functionality — strengthening code quality through hands on debugging, testing, and optimization along the way.
- Operated within an Agile workflow — using Git/GitHub feature branches, pull requests, and peer code review to consistently deliver sprint commitments.

AI & Automation Engineer

Apr 2025 – Sep 2025

Armstrong — Alexandria, Egypt

- Engineered AI-driven automation pipelines for the Digital Egypt Cubs Initiative (DECI), a national educational initiative under Egypt's Ministry of Communications and Information Technology (MCIT).
- Architected scalable content-generation workflows using n8n, custom GPTs, LLM systems, and open-source AI models, automating educational content production end-to-end.
- Cut manual production workload by shifting to an automation-first pipeline design, combining AI-assisted content generation with orchestrated visual-asset workflows.

PROJECTS

Mind Guard — AI-Powered Mental Health Monitoring Ecosystem

Graduation Project

Tech: Python, LangChain, LangGraph, ChromaDB, BM25, Google Gemini API, HRV/PPG Signal Processing, Next.js, NestJS

- Co-developed an end-to-end AI mental-health ecosystem integrating wearable HRV/PPG signals, behavioral data, clinical screening scales, and AI-generated risk insights into a unified doctor-patient monitoring workflow.
- Built a deterministic PPG signal-processing pipeline — filtering, normalization, heartbeat peak detection, RR interval construction, and HRV feature extraction — converting raw wearable data into clinically meaningful metrics.
- Designed NEURA, a safety-aware RAG chatbot with crisis detection, guardrails, language detection, and query rewriting, enabling reliable, context-grounded mental-health conversations.
- Implemented a hybrid retrieval system combining ChromaDB dense search, BM25 sparse retrieval, and Reciprocal Rank Fusion with re-ranking to improve the relevance of retrieved context.

VisionDeck — Computer Vision Suite

Tech: Python, OpenCV, MediaPipe, YOLOv8, Streamlit

- Engineered a Streamlit-based computer vision dashboard integrating face detection, hand-gesture recognition, and real time object detection (YOLOv8), with support for both webcam and IP camera video streams.

Robotics Control Systems — Biped Robot & Robotic Arm (DUM-E)

Tech: Python, Arduino/C++, ESP32, Streamlit, Computer Vision, Forward/Inverse Kinematics

- Engineered the control system for a six-servo, ESP32 powered biped robot, building a Streamlit desktop application for real-time diagnostics, pose editing, joint calibration, and color-based visual tracking for autonomous forward locomotion.
- Built a separate robotic-arm control stack (DUM-E) — firmware, endpoint discovery, and forward/inverse kinematics helpers — with sequence replay and guarded pick-and-place execution to prevent collisions during autonomous operation.

KBAI — Course Knowledge Chatbot

Tech: Python, Google Gemini API, Retrieval-Augmented Generation (RAG), NLP

- Built a retrieval augmented chatbot using the Gemini API, grounding responses in retrieved course material to generate accurate, explainable answers and reduce hallucinated output.

Additional AI & Algorithms Projects

Tech: Python, Java, JavaFX, NLP, Minimax, Alpha-Beta Pruning, A*, RNN/Transformer Architectures

- Neural Machine Translation (English–Spanish): built a translation pipeline — parallel-corpus cleaning, tokenization, and sequence-to-sequence/Transformer-based modeling — evaluated using BLEU scores.
- Search, Optimization & Games: implemented a Connect 4 AI, 8-Puzzle solver, and TSP-style route optimizer using Minimax with Alpha-Beta pruning, A*, UCS, and Greedy Best-First Search.

LEADERSHIP EXPERIENCE

Head of Media

ElCoder — Alexandria, Egypt

- Led the Media department at ElCoder, directing event coverage, social media strategy, visual branding, and promotional content for workshops and student activities.
- Coordinated photo and video documentation, content calendars, and campaign assets, strengthening audience engagement and the organization's public presence.

EDUCATION

Bachelor's Degree in Computers and Data Science — Artificial Intelligence Department

Alexandria University — Graduated 2026

Relevant Coursework: Machine Learning, Natural Language Processing, Pattern Recognition, Neural Network Databases, Cloud Computing, Operations Research, Embedded Systems, Modern Control Systems, Internet of Things, Web Development
